

ERFURT WEIMAR, Germany

WMO: 105540

Lat: 50.9828N

Lon: 10.9608E

Elev: 1056

StdP: 14.14

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	15.8	6.7	8.1	13.2	11.8	10.4	17.6	32.3	45.8	28.7	42.8	7.1	210	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	86.2	66.6	82.4	65.4	78.8	64.4	68.4	80.9	67.0	79.2	65.5	76.4	8.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	94.6	71.6	62.9	89.1	70.3	61.2	84.0	69.1	33.2	81.5	31.9	79.0	30.8	76.2	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	21.3	18.8	DB	6.2	92.9	6.3	2.6	1.6	94.8	-2.1	96.4	-5.6	97.8	-10.2	99.8	(4)
(5)			WB	5.7	72.4	6.2	1.9	1.2	73.7	-2.4	74.8	-5.9	75.9	-10.5	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	31.9	32.9	39.8	49.0	55.1	61.8	65.9	64.3	58.1	49.7	41.2	35.7	(6)
(7)		DBStd	13.93	9.08	8.72	8.14	7.39	7.29	6.52	6.30	6.07	6.08	7.10	7.27	7.97	(7)
(8)		HDD50	2323	560	479	325	106	32	2	0	0	4	95	277	443	(8)
(9)		HDD65	6110	1025	899	782	480	316	136	66	87	222	476	714	907	(9)
(10)		CDD50	1915	0	0	8	76	191	356	494	443	248	85	13	1	(10)
(11)		CDD65	229	0	0	0	1	11	40	95	65	16	1	0	0	(11)
(12)		CDH74	2102	0	0	0	11	96	373	890	589	139	4	0	0	(12)
(13)		CDH80	606	0	0	0	1	14	90	300	179	22	0	0	0	(13)
(14)	Wind	WSAvg	9.0	10.8	9.8	9.9	8.3	8.2	7.7	8.1	7.8	8.2	8.6	9.5	11.0	(14)
(15)	Precipitation	PrecAvg	20.3	1.0	0.9	1.4	1.6	2.4	2.4	2.5	2.3	1.7	1.5	1.5	1.3	(15)
(16)		PrecMax	30.1	2.9	2.1	3.8	5.3	7.6	4.5	8.0	5.7	4.7	4.4	4.1	3.1	(16)
(17)		PrecMin	11.4	0.0	0.0	0.2	0.3	0.4	0.2	0.4	0.6	0.4	0.1	0.0	0.0	(17)
(18)		PrecStd	4.3	0.5	0.6	0.8	0.9	1.3	1.0	1.4	1.3	0.9	1.0	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	55.1	64.8	75.1	82.0	87.8	92.9	91.1	82.6	73.5	61.1	54.0	(19)
MCWB			49.6	45.0	51.2	57.6	65.4	67.6	67.9	66.6	63.6	60.5	55.5	49.0	(20)	
2%		DB	50.0	50.1	59.3	70.0	76.7	82.3	87.5	84.6	78.5	66.6	57.3	51.6	(21)	
		MCWB	46.1	44.9	49.5	56.1	63.5	65.9	66.3	66.1	62.6	58.7	52.7	47.6	(22)	
5%		DB	46.7	46.7	55.3	66.4	71.9	78.5	82.7	80.5	73.5	62.9	55.0	48.4	(23)	
		MCWB	44.1	42.9	47.9	54.5	59.9	65.1	65.6	65.3	61.3	56.4	50.4	44.9	(24)	
10%		DB	44.3	44.4	51.6	62.5	68.2	74.3	78.8	76.6	69.5	60.5	51.8	46.1	(25)	
		MCWB	42.1	40.9	45.8	52.8	58.0	63.2	64.2	62.9	59.9	55.3	48.0	43.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	48.3	53.8	60.3	68.0	71.1	70.0	66.5	61.5	55.8	50.3	(27)	
MCD B			53.3	50.7	61.4	71.4	77.7	82.1	83.2	82.1	78.5	69.2	60.1	53.5	(28)	
2%		WB	46.7	45.6	50.8	57.4	64.4	67.9	68.1	67.8	64.3	59.3	53.0	48.1	(29)	
		MCD B	49.7	49.3	57.4	67.5	73.8	79.4	81.5	80.8	74.6	66.3	56.9	51.0	(30)	
5%		WB	44.2	43.1	48.7	55.5	61.7	65.6	66.9	66.2	62.2	57.3	50.6	45.3	(31)	
		MCD B	46.6	46.7	54.3	64.8	69.9	75.7	79.4	78.2	71.6	62.5	54.2	48.3	(32)	
10%		WB	41.5	40.6	46.3	53.6	59.1	63.7	65.4	64.6	60.5	55.3	48.5	43.2	(33)	
		MCD B	43.4	43.5	51.4	61.6	66.9	72.7	76.5	73.9	67.5	59.8	51.6	45.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	10.2	13.8	17.6	17.0	17.6	18.3	17.9	16.2	12.7	9.0	8.2	(35)
MCDBR			9.9	14.2	21.5	24.1	22.1	23.8	25.2	24.7	23.0	17.4	12.0	10.8	(36)	
MCWBR			8.1	10.2	13.1	13.2	12.0	11.8	9.3	10.2	10.8	10.6	8.8	8.5	(37)	
5% WB		MCDBR	9.5	13.2	18.8	21.6	20.6	22.1	21.1	20.9	19.1	16.2	10.9	9.9	(38)	
		MCWBR	8.0	10.0	12.3	12.3	11.8	11.6	8.8	9.7	10.1	10.4	8.6	8.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.326	0.364	0.403	0.407	0.404	0.419	0.412	0.377	0.350	0.327	0.299	(40)	
taud		2.408	2.368	2.293	2.224	2.243	2.272	2.235	2.277	2.358	2.425	2.428	2.401	(41)		
Ebn at Noon		230	254	263	265	269	271	264	259	256	242	218	212	(42)		
Edh at Noon		19	26	34	40	41	41	42	38	31	24	19	17	(43)		
(44)	All-Sky Solar Radiation	RadAvg	239	464	805	1282	1544	1677	1583	1361	981	571	267	180	(44)	
(45)		RadStd	37	67	103	161	180	141	199	136	133	68	36	22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page